

Exploratory Data Analysis Report

Dataset: Titanic Dataset

Tools Used: Python (Pandas, Matplotlib, Seaborn)

1. Data Overview

Basic Info

- Rows: 891
- Columns: 12
- Key Features: Name, Age, Sex, Fare, Pclass, Survived, Embarked, etc.

Missing Values

- Age: 177 missing
- Cabin: 687 missing
- Embarked: 2 missing

Actions Taken

- Used .info(), .describe(), .value_counts()
- Removed or imputed missing values where necessary

2. Univariate Analysis

Gender Distribution

- More males than females
- Sex: ~65% Male, ~35% Female

Survival Distribution

- 62% passengers died
- 38% survived

Age Distribution

- Most passengers between 20-40 years
- Some infants and elders too

Class Distribution

- 3rd class had the most passengers

3. Bivariate Analysis

Survival by Gender

- Females had a much higher survival rate
- Visual: Countplot showed higher female survival

Survival by Passenger Class

- 1st Class passengers had the highest survival rate
- 3rd Class had the lowest survival rate

Age vs Survival

- Many younger passengers survived
- Older passengers had lower survival chances

4. Correlation Matrix

Feature	High Correlation With
Fare	Pclass (negative)
Survived	Pclass, Fare, Sex

- Heatmap showed Pclass, Fare, and Sex strongly correlate with survival

5. Visuals Used

- **Countplots:** For gender, survival, class distribution
- **Histograms:** For age, fare
- **Boxplots:** Age vs Class
- **Heatmap:** Correlation matrix
- **Pairplot:** To explore numeric relationships

6. Key Insights

-  **Gender played a major role** – women had better chances of survival
-  **Higher class = higher survival** – 1st Class passengers were safest
-  **Young passengers** had higher survival
-  Fare amount positively correlated with survival
-  Embarked location slightly influenced survival